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RESEARCH ARTICLE



Associations Between Personality Traits and ChatGPT Usage: The Dual Mediating Roles of Social Image and Computer Self-Efficacy

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ABSTRACT

ChatGPT has emerged as a focal point of interest among researchers and practitioners. However, little empirical research has examined its usage. Based on the Five Factor Model and the Technology Acceptance Model, this research examines the relationships between personality traits, specifically extraversion, openness, and conscientiousness, and ChatGPT usage, with social image and computer self-efficacy serving as mediators. In March 2023, this study surveyed 784 users in China, consisting of 462 males and 322 females, with an average age of 23.7. Our analysis reveals that extraversion, openness, and conscientiousness are positively associated with ChatGPT usage through sequential mediating involving social image and computer self-efficacy. While these traits demonstrate indirect associations with usage, openness and conscientiousness lack direct relationships with ChatGPT usage, and neither extraversion nor openness directly relates to computer self-efficacy. These findings offer in-depth perspectives for generative artificial intelligence technology developers and marketers to enhance user engagement strategies at their early stages.

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Five factor model;
technology acceptance
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extraversion; openness;
conscientiousness

Introduction

The rapid advancement of artificial intelligence (AI), particularly with the development of the Generative Pre-trained Transformer (ChatGPT) by OpenAI, marks a significant milestone in AI-driven technologies. ChatGPT is a state-of-the-art language model that uses advanced deep learning techniques to process and generate human-like text based on input prompts (Ma & Huo, 2023). Its remarkable capabilities render it an indispensable tool across various industries, including healthcare, teaching, business operations, and beyond, highlighting its versatility and potential to revolutionize various aspects of society (Raj et al., 2023). Given the recent emergence of ChatGPT, limited research has examined its usage patterns (Ma et al., 2025). However, understanding the usage of ChatGPT is crucial because the benefits and impacts of information technology depend on how individuals integrate these technologies into their work

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activities. Considering that impressions may differ greatly based on prior experiences with various AI products, this study concentrates exclusively on ChatGPT rather than multiple AI platforms. Therefore, this study aims to investigate the relationships between personality traits, namely extraversion, openness, and conscientiousness, and ChatGPT usage, examining the mediating effects of social image and computer self-efficacy.

Literature Review and Hypotheses

Theoretical Background

This study integrates the Five Factor Model and Technology Acceptance Model 3 (TAM3) as its theoretical foundation. The Five Factor Model was selected over alternative frameworks (HEXACO, Myers-Briggs Type Indicator, PEN Model) for its comprehensive dimensions and broad applicability (Rivers, 2021). Similarly, TAM3 was chosen among competing models (MPCU, UTAUT, DeLone and McLean's Information Systems Success Model) for its comprehensive cross-domain applicability (Mehra et al., 2021). Furthermore, the relationship between social image and computer self-efficacy remains underexamined despite their significance in user behavior. Therefore, this research employs TAM3 with specific focus on these two constructs to provide targeted insights. The integrated framework addresses both internal drivers through personality traits and external factors through situational perceptions in ChatGPT usage, creating a complementary analytical approach where the Five Factor Model identifies potential users through intrinsic traits while TAM3 explains usage mechanisms.

Big Five Personality Traits and ChatGPT Usage

Personality traits, as enduring behavioral patterns (McCrae & Costa, 1999), predict various social behaviors including technology usage (Abdelrahman, 2022). The Five Factor Model distinguishes five fundamental personality dimensions: extraversion, openness, conscientiousness, agreeableness, and neuroticism. The present study examines relationships between three specific personality traits and ChatGPT usage: extraversion, openness, and conscientiousness. Extraversion encompasses interpersonal warmth, sociability, and energy-seeking, reflecting tendencies to engage actively with others. Openness manifests as receptivity to novel ideas, creativity, and esthetic appreciation, capturing inclinations toward exploration and intellectual diversity. Conscientiousness characterizes organization, goal-orientation, and self-discipline, demonstrating preferences for structured objective pursuit. Based on TAM3, we conceptualize usage behavior as the degree of active ChatGPT engagement in daily tasks (Venkatesh & Bala, 2008). Given ChatGPT represents an emerging AI technology, limited research has examined relationships between these personality traits and ChatGPT usage. Therefore, this study investigates associations between these three personality factors and ChatGPT usage.

Our study examines extraversion, openness, and conscientiousness in relation to ChatGPT usage, excluding agreeableness and neuroticism based on theoretical and empirical considerations. The literature presents mixed findings regarding agreeableness and neuroticism in technology usage, with some studies showing positive associations while others reveal no significant relationships (Denden et al., 2022; Gupta & Bhaskar,

2023; Pinho et al., 2020). In contrast, extraversion, openness, and conscientiousness consistently demonstrate central roles in information technology usage, especially among early users (Reith et al., 2020; Seibert et al., 2021). Innovation diffusion theory (IDT) identifies five user categories based on the timing of innovativeness: innovators, early adopters, early majority, late majority, and laggards (Rogers, 2003). With ChatGPT launched in November 2022 and data collected in March 2023, early users comprise the sample population of our study. As prior studies often do not distinguish innovators from early adopters, they are treated as a single group. Rogers describes early users as innovative, proactive opinion leaders, aligning with extraversion, openness, and conscientiousness, which reflect novelty seeking, forward thinking, and goal-directed behavior (Hilal & Varela-Neira, 2022). In contrast, agreeableness and neuroticism demonstrate stronger alignment with late majority and laggard characteristics. Highly agreeable individuals prioritize interpersonal harmony and trust technologies after widespread implementation, mirroring late majority behavior. Similarly, individuals with higher scores in neuroticism often overestimate technological risks, aligning with laggards who resist innovation (Costa & McCrae, 1992a; 1992b; Rogers, 2003). This theoretical alignment supports the focus on extraversion, openness, and conscientiousness when investigating ChatGPT usage among early users, allowing for targeted analysis and reduced model complexity.

Personality traits, such as extraversion, openness, and conscientiousness, may be positively associated with ChatGPT usage behavior. The link between personality traits and technology usage behavior also gains support from the Five Factor Model, which postulates that different personality traits are linked to cognitive processes, emotional responses, and behavioral patterns (McCrae & Costa, 1999). To be specific, individuals with high extraversion levels frequently engage in social interactions and may utilize ChatGPT for simulated conversations, especially when direct social engagement is unavailable (Sindermann et al., 2020). Moreover, individuals with high openness traits demonstrate an inclination toward novel experiences and may incorporate ChatGPT to expand their knowledge horizons (Aydın, 2019; Lissitsa & Kol, 2021). Additionally, individuals with high conscientiousness are ambitious and have a strong drive to achieve success. They may employ ChatGPT to access information and enhance productivity, effectively achieving their goals (Gupta & Bhaskar, 2023). Given the theoretical and empirical evidence, we propose the following hypotheses regarding the direct effects of personality traits on ChatGPT usage:

H1a: Extraversion is positively associated with ChatGPT usage behavior.

H1b: Openness is positively associated with ChatGPT usage behavior.

H1c: Conscientiousness is positively associated with ChatGPT usage behavior.

Social Image as the Mediator

Social image encompasses two primary categories across disciplines. The first, common in psychology, sociology, and marketing, views it as external perceptions shaped by media, culture, and collective behavior, such as how others perceive a person (Rodríguez Mosquera et al., 2011) or a brand (Keller, 1993). The second, rooted in cognitive

psychology, views social image as individual self-perception of how others perceive me (Markus & Wurf, 1987). This study applies TAM3 to examine ChatGPT usage behavior, defining social image as the degree to which ChatGPT usage is perceived to enhance individual status within a social system (Moore & Benbasat, 1991), aligning with cognitive psychology perspectives. A higher social image means people believe that using ChatGPT elevates their standing, while a lower social image suggests neutrality or potential loss. Similar constructs exist across technology usage models, including subjective norm (TRA/TPB/TAM2/TAM3), social influence (UTAUT), and social factors (MPCU), all sharing the foundation that anticipated social evaluation of technology usage is associated with individual behavior. However, social image specifically focuses on individual perceptions of status enhancement, while subjective norm emphasizes interpersonal pressure and social factors address external group influences (Thompson et al., 1991; Venkatesh et al., 2003). TAM3 theorizes that subjective norm is positively related to social image, as performing behaviors endorsed by important social group members is perceived to enhance social standing (Venkatesh & Bala, 2008). Social image is relative and dynamic, reflecting group-specific values within particular contexts rather than absolute measures, evolving as societal priorities change (Moore & Benbasat, 1991). Social image is a pivotal factor in investigating ChatGPT usage behavior, as TAM3 indicates that social image shows a significant association with technology usage behavior. Also, face-saving and group conformity are fundamental in Chinese culture, making social image essential for understanding ChatGPT usage in Chinese contexts (Zhang et al., 2020). However, research examining the relationship between personality traits and social image remains limited. Further, as ChatGPT represents a new technological tool, studies examining the connection between social image and the usage of ChatGPT are equally scarce. Therefore, this study examines social image as a mediator between personality traits and ChatGPT usage.

Personality characteristics, such as extraversion, openness, and conscientiousness, may be positively associated with social image. In detail, extraverted individuals engage actively in group activities and seek recognition within their social networks. These individuals gravitate toward novel technologies to demonstrate technological proficiency, thereby gaining peer admiration. Thus, individuals high in extraversion use ChatGPT as it is perceived to enhance their social status (Hilal & Varela-Neira, 2022). Similarly, individuals high in openness are creative and enjoy using the latest technology to create and innovate. Therefore, those with high levels of openness use ChatGPT as it is perceived to enhance their social standing (Lai, 2019). Furthermore, individuals who are highly conscientious place significant value on skill development and problem-solving capabilities. They appreciate using the latest technology to enhance their problem-solving skills. Consequently, highly conscientious individuals use ChatGPT as it is perceived to elevate their social status (Wang et al., 2022).

Social image, in turn, is expected to play a pivotal role in increasing ChatGPT usage behavior. Grounded in TAM3, social influence relates to individual behavior through three mechanisms: compliance, internalization, and identification (Venkatesh & Davis, 2000). Within this framework, social image enhances perceived usefulness through identification processes. When individuals perceive that using ChatGPT enhances their social image, they evaluate it as more useful because ChatGPT enables them to align with socially valued groups. The identification process occurs as

individuals align their behavior with group norms to derive a sense of belonging. Thus, the perceived usefulness of ChatGPT stems not only from its practical utility but also from its symbolic value, which subsequently drives its usage behavior.

Given that these personality traits may be positively associated with social image, and recognizing the association between social image and ChatGPT usage behavior, we propose the following hypotheses:

H2a: Social image mediates the link between extraversion and ChatGPT usage behavior.

H2b: Social image mediates the link between openness and ChatGPT usage behavior.

H2c: Social image mediates the link between conscientiousness and ChatGPT usage behavior.

Computer Self-Efficacy as the Mediator

Drawing from TAM3, computer self-efficacy is defined as the extent to which individuals believe in their capability to execute specific tasks using ChatGPT, showing a significant association with perceived ease of use (Venkatesh & Bala, 2008). Several established technology usage models include related constructs, such as perceived behavioral control in TPB. While both assess perceived capability, perceived behavioral control also considers external factors like resources and barriers (Ajzen, 1991). TAM3 proposes that computer self-efficacy demonstrates a strong relationship with technology usage behavior (Venkatesh & Bala, 2008). Despite this significance, limited empirical research examines how personality characteristics are associated with computer self-efficacy (Manolika et al., 2022). Therefore, this study endeavors to scrutinize the mediating role of computer self-efficacy between Big Five personality traits and ChatGPT usage.

Personality attributes, such as extraversion, openness, and conscientiousness, may be positively related to computer self-efficacy. In detail, extraversion enhances computer self-efficacy because extraverts consistently demonstrate positive emotions and attitudes, and their optimism strengthens their belief in utilizing resources to overcome technological challenges (San-Martin et al., 2020). Furthermore, highly open-minded individuals exhibit enhanced computer self-efficacy in ChatGPT utilization, as their intellectual curiosity motivates active exploration and familiarization with diverse ChatGPT functions and methods, thereby increasing their confidence (Zheng et al., 2020). Similarly, individuals with high conscientiousness display improved computer self-efficacy in ChatGPT usage because their structured and methodical approach to learning facilitates a comprehensive understanding of the features of ChatGPT, thereby fostering a sense of confidence (Rivers, 2021).

Moreover, computer self-efficacy could be linked to increased ChatGPT usage. The TAM3 suggests that higher computer self-efficacy associates with perceived ease of use, leading to a greater likelihood of using new technologies (Agarwal et al., 2000; Venkatesh & Bala, 2008). Enhanced confidence in ChatGPT utilization diminishes perceived difficulty. When individuals perceive ChatGPT as straightforward to operate and requiring minimal cognitive effort to master its functionalities, rapid learning is enabled. Consequently, this ease of use makes ChatGPT more appealing for both

occasional and frequent usage. Empirical evidence also supports this view. For example, studies have demonstrated that individuals possessing confidence in their interactions with information technology exhibit greater involvement and effectiveness in its use (Maican et al., 2019).

Considering that these personality traits may be positively associated with computer self-efficacy, and given the connection between computer self-efficacy and ChatGPT usage behavior, we propose the following hypotheses:

H3a: Computer self-efficacy mediates the link between extraversion and ChatGPT usage behavior.

H3b: Computer self-efficacy mediates the link between openness and ChatGPT usage behavior.

H3c: Computer self-efficacy mediates the link between conscientiousness and ChatGPT usage behavior.

Serial Mediation Model

Social image may be positively related to computer self-efficacy. The rationale behind this assertion lies in social cognitive theory (SCT), which posits that self-efficacy is bolstered by four principal sources: personal experiences, vicarious experiences *via* observation of others, verbal persuasion, and physiological feedback (Bandura, 1977). Encouragement of use represents verbal persuasion, a key source of self-efficacy (Bandura, 1986). According to TAM3, subjective norm is positively associated with social image, reflecting how social pressure from important others is linked to perception (Venkatesh & Bala, 2008). When significant social group members advocate for ChatGPT usage, the individual may perceive that using ChatGPT would enhance their standing within the group (Venkatesh & Davis, 2000). Consequently, social image, which represents receiving supportive encouragement from a social reference group (Chahal & Rani, 2022), may enhance the confidence of individuals in effectively utilizing ChatGPT. Prior studies demonstrate that social persuasion constitutes a strong factor in enhancing computer self-efficacy across various contexts, including social media usage and computer applications for pre-service teachers (Chahal & Rani, 2022; Chibisa et al., 2021). Concurrently, SCT posits a positive relationship between environmental and personal factors (Bandura, 1986), suggesting that social image functions as a crucial environmental factor potentially associated with computer self-efficacy. In other words, computer self-efficacy, individual beliefs about their technological capabilities, may not be realistic without sufficient extrinsic information about the behavior.

Given the evidence suggesting that social image has a positive association with computer self-efficacy and the mediating role of social image and computer self-efficacy between personality traits and ChatGPT usage behavior, we propose the following hypotheses:

H4a: Social image is positively associated with computer self-efficacy.

H4b: Social image and computer self-efficacy sequentially mediate the association of extraversion with ChatGPT usage behavior.

H4c: Social image and computer self-efficacy sequentially mediate the association of openness with ChatGPT usage behavior.

H4d: Social image and computer self-efficacy sequentially mediate the association of conscientiousness with ChatGPT usage behavior.

These hypotheses were not pre-registered.

Method

Participants and Procedure

Following the Declaration of Helsinki, data was collected through an online survey in March 2023 *via* popular social media platforms, including Weibo, WeChat, and Douban. Trained graduate students were responsible for data collection. Prior to participation, respondents were notified of the voluntary nature of the study and their right to withdraw from the survey at any time. To encourage participation, monetary incentive of 5 RMB was provided to each respondent. The survey was limited to individuals with prior ChatGPT experience. Through convenience sampling, 949 responses were collected. Ultimately, after excluding responses that failed attention check questions and showed consistent response patterns throughout the questionnaire, 784 valid responses were retained for analysis. This sample size exceeded both the optimal sample size determined by a priori power analysis (G*Power: F tests - Linear multiple regression: Fixed model, R^2 deviation from zero; $f^2 = 0.15$, $\alpha = 0.05$, $\beta = 0.80$, $n = 77$; Faul et al., 2009) and the recommended sample size for cross-sectional questionnaire surveys ($Z = 1.96$, $E = 0.05$, $p = 0.60$, $n = 368$; Chow et al., 2017), ensuring adequacy for structural equation modeling (SEM). The demographic profile of the respondents was analyzed and summarized in Table 1. It is observed that a relatively balanced gender distribution with slightly more male participants. Most respondents were aged 19–29 years, resided in eastern coastal cities, and held bachelor's degrees or higher. The sample included both students and working professionals, providing comprehensive representation.

Measures

The questionnaire comprised two sections: demographics and study variables. The instrument included 19 items measuring six factors: extraversion, openness, conscientiousness, social image, computer self-efficacy, and ChatGPT usage. Personality measures were adapted from Costa and McCrae (1992b), while other items were derived from Venkatesh and Bala (2008) and Moon and Kim (2001). All items used 5-point Likert scales (1 = strongly disagree, 5 = strongly agree), except for one ChatGPT usage item which employed ordinal multiple-choice options. Specific items of each construct are presented in Table 2.

Although the original measurement items were developed in English, the Chinese version demonstrates strong reliability and validity through comprehensive validation procedures. At first, the translation process involved rigorous verification by three bilingual experts proficient in English and Chinese to ensure linguistic and conceptual equivalence between the two versions. The measurement items were then carefully

Table 1. Demographic Profiles of Participants (*N*=784).

Characteristics	Frequency (Percentage)
Gender	
Male	462 (58.93%)
Female	322 (41.07%)
Age	
18 years old and below	20 (2.55%)
19–29	728 (92.86%)
30–40	35 (4.46%)
Above 40 years old	1 (0.13%)
Geographical area	
Eastern China	452 (57.65%)
Central China	174 (22.19%)
Western China	127 (16.2%)
Northeastern China	20 (2.55%)
Hong Kong, Macao, and Taiwan Regions of Cina	3 (0.38%)
Overseas	8 (1.02%)
Educational background	
Below undergraduate	47 (5.99%)
Undergraduate	540 (68.88%)
Graduate and above	197 (25.13%)
Role in Society	
Students	556 (70.92%)
Employed Individuals	228 (29.08%)

modified to fit the context of this study, such as changing “system” to “ChatGPT.” In addition, the Five Factor Model and TAM3 have been rigorously validated across various cultures, including China. Therefore, some measurement items used in this study were derived from prior Chinese validation studies (Hong et al., 2021; Li, 2013; Zhu et al., 2022), ensuring scientific rigor and high measurement quality. Also, to ensure instrument clarity and reliability, a pilot study was conducted with eight participants. Based on their feedback, we revised several items to reduce ambiguity before distributing the final survey.

Statistical Analysis

The data analysis process consisted of multiple stages utilizing statistical software packages. Initial data processing and cleaning were conducted using SPSS 26.0. The questionnaire included reverse-coded items to mitigate response bias that might arise from participants responding habitually to consecutive positively worded items. These items were reverse coded to ensure consistency in scoring directions across all questionnaire items, thereby enabling harmonized data for subsequent statistical analyses.

Following this, SEM was developed to examine the hypothesized associations among variables. This analytical approach was selected because the study incorporated multiple independent and mediators, and SEM enables simultaneous analysis of multiple variables (Hair et al., 1998). Moreover, all the core variables in this study were latent constructs based on multiple items, and SEM facilitates the analysis of relationships between latent variables measured by multiple indicators (DeVellis & Thorpe, 2021). The analysis began with the verification of whether the nineteen measurements appropriately reflected the six latent variables in the measurement model. Firstly, as a prerequisite for model validity, Cronbach’s Alpha (CA) values were calculated for each construct using SPSS 26.0 to assess reliability, examining the consistency and stability

Table 2. Confirmatory Factor Analysis of Key Variables ($N=784$).

Constructs and items	Standardized factor loadings	SMC	CA	CR	AVE
Extraversion			0.748	0.763	0.525
I get bored with crowded parties.	0.821	0.674			
I prefer to keep to myself rather than being around others.	0.780	0.608			
I enjoy participating in social and entertainment gatherings	0.540	0.292			
Openness			0.810	0.821	0.607
I am open to experiencing new things	0.855	0.731			
I often seek out new activities that break from routine.	0.811	0.658			
I am curious about different ideas and approaches to life.	0.657	0.432			
Conscientiousness			0.683	0.687	0.423
I think carefully before making decisions or taking action.	0.594	0.353			
I set high standards for myself and work hard to achieve them.	0.659	0.434			
I make an effort to plan things out in an orderly manner.	0.695	0.483			
Social image			0.906	0.901	0.696
Because I use ChatGPT, others may think I have a higher status.	0.882	0.778			
Because I use ChatGPT, others may think I have a better reputation compared to those who do not use it.	0.913	0.833			
Because I use ChatGPT, others may think I have a more positive image among my peers.	0.724	0.524			
Because I use ChatGPT, others may think I have a higher profile.	0.805	0.648			
Computer self-efficacy			0.739	0.740	0.489
Even without guidance, I believe I can utilize ChatGPT effectively.	0.680	0.463			
I feel confident in using ChatGPT to complete tasks.	0.779	0.607			
I could effectively use ChatGPT with the built-in tools.	0.630	0.396			
Usage behavior			0.757	0.761	0.517
How long have you been using ChatGPT	0.724	0.524			
How many times do you use ChatGPT per week	0.562	0.388			
On average, how much time do you spend using ChatGPT per week	0.800	0.640			

SMC: Squared multiple correlations; CA: Cronbach's alpha; CR: Composite reliability; AVE: Average variance extracted.

among different items within a single latent variable (Nunnally & Bernstein, 1994). Secondly, standardized factor loadings were computed through confirmatory factor analysis (CFA), which generated squared multiple correlations (SMC), composite reliability (CR), and average variance extracted (AVE). These indicators were essential for assessing convergent validity, determining whether items within a single latent variable jointly measured the latent variable (Bagozzi & Yi, 1988; Fornell & Larcker, 1981). Thirdly, we compared the relationships between constructs computed through SPSS 26.0 with the square roots of AVE in the discriminant analysis, in order to verify that

the measurement model assessed distinct constructs (Fornell & Larcker, 1981). Fourthly, CFA employed multiple indices to assess the overall fit between the measurement model and the data, including chi-square/degrees of freedom ratio (χ^2/Df), goodness-of-fit index (GFI), among others (Bentler, 1990; Hu & Bentler, 1999). Fifthly, the maximum likelihood method was employed to calculate path coefficients and associated p-values in the structural model. Standardized path coefficients (β) were utilized to facilitate comparisons across different variables within the model, with the magnitude of coefficients indicating relationship strength and p-values determining their significance levels (Kline, 2016).

Subsequently, the bootstrapping method assessed the direct and indirect effects of three personality traits on ChatGPT usage through two mediators. The bias-corrected bootstrap method generated standard errors and confidence intervals for the estimates. The analysis utilized 5000 bootstrap samples to obtain bias-corrected bootstrap with 95% confidence intervals (Preacher & Hayes, 2008). While SPSS 26.0 was employed for data cleaning, reliability testing, and correlation analyses, AMOS 28.0 performed all other analytical procedures, as these tools are widely utilized in statistical analyses.

Results

Measurement Model Testing

Table 2 presents the values for reliability and convergent validity of the measurement model. The CA values for all items exceeded 0.60, with the majority surpassing 0.70, demonstrating acceptable reliability among the items and indicating high internal consistency within each construct (Li et al., 2023; Mehra et al., 2021; Nunnally & Bernstein, 1994; Tur-Porcar et al., 2024). In examining the factor structure, all items demonstrated strong loadings above 0.50, with a substantial proportion reaching beyond 0.60, indicating each item loaded significantly on its underlying construct (Hair et al., 1998). All the SMCs surpassed the threshold of 0.25, with many items achieving scores above 0.36 (Bagozzi & Yi, 1988). The CR values for all items exceeded 0.60, with the majority surpassing 0.70. The AVE value revealed satisfactory levels above the 0.40 threshold, and a substantial portion exceeded 0.50 (Fornell & Larcker, 1981; Gebremedhin et al., 2022; Lam, 2012). These indicators of factor loadings, SMC, CR, and AVE values collectively suggest an acceptable level of convergent validity across all constructs, indicating that all items within individual latent variables effectively measure their respective constructs.

Table 3 illustrates that all square roots of the AVE values in the diagonal of the matrix exceed the corresponding inter-construct relationships, thereby substantiating the discriminant validity of the scales employed, indicating that different latent variables are independent of each other (Fornell & Larcker, 1981).

The fit indices of the measurement model analysis presented in Table 4 indicate that all values substantially exceeded their respective criteria levels. The comprehensive assessment suggests that the measurement model demonstrates a good fit with the collected data, validating the applicability of the measurement model in this study. Therefore, it is appropriate to establish the structural model to examine the

Table 3. Discriminant Validity.

	EXV	OP	CST	SIM	CSE	UB
EXV	0.724					
OP	0.290***	0.779				
CST	0.275***	0.366***	0.650			
SIM	0.265***	0.322***	0.361***	0.841		
CSE	0.199***	0.284***	0.401***	0.292***	0.700	
UB	0.341***	0.222***	0.332***	0.423***	0.450***	0.718

EXV: Extraversion; OP: Openness; CST: Conscientiousness; SIM: Social image; CSE: Computer self-efficacy; UB: Usage behavior.

* $p < .05$, ** $p < .01$, *** $p < .001$.

Table 4. Fit Indices for the Measurement Model.

Model	χ^2	Df	χ^2/Df	GFI	RMSEA	CFI	NFI
Criteria			<3	≥ 0.90	<0.10	≥ 0.90	≥ 0.90
Obtained	375.878	134	2.805	0.953	0.048	0.961	0.941

Df: Degrees of freedom; GFI: Goodness-of-fit index; RMSEA: Root mean square error of approximation; CFI: Comparative fit index; NFI: Normed fit index. Cutoff values for model fit indices are based on the recommendations by Hu and Bentler (1999), and Bentler (1990).

relationships between the three personality traits, social image, computer self-efficacy, and ChatGPT usage through path coefficients.

Structural Model Testing

Table 5 presents the evaluation of standardized path coefficients and their significance levels through p-values. Path coefficients were determined to be statistically significant when p-values were less than 0.05 (Kline, 2016). The analysis reveals several significant relationships among the variables. Extraversion exhibited positive associations with social image ($\beta = 0.119$, $p = 0.005$) and ChatGPT usage behavior ($\beta = 0.279$, $p < 0.001$). Openness demonstrated positive relationships with social image ($\beta = 0.166$, $p < 0.001$). Conscientiousness showed positive relationships with social image ($\beta = 0.318$, $p < 0.001$) and computer self-efficacy ($\beta = 0.480$, $p < 0.001$). Moreover, both social image ($\beta = 0.336$, $p < 0.001$) and computer self-efficacy ($\beta = 0.319$, $p < 0.001$) exhibited positive relationships with ChatGPT usage behavior. The social image also demonstrated a positive relationship with computer self-efficacy ($\beta = 0.110$, $p = 0.015$).

In summary, the analyses revealed statistically significant effects along all paths except for four: neither extraversion nor openness is positively related to computer self-efficacy, and neither openness nor conscientiousness is positively related to ChatGPT usage behavior. Consequently, our study found hypotheses H1b and H1c were not supported, while providing empirical support for hypotheses H1a and H4a. Figure 1 provides a visual representation of the relationships among variables and displays the standardized coefficients for each proposed path in the study, with significant relationships depicted by solid lines and non-significant relationships indicated by dashed lines.

Serial Mediation Model Testing

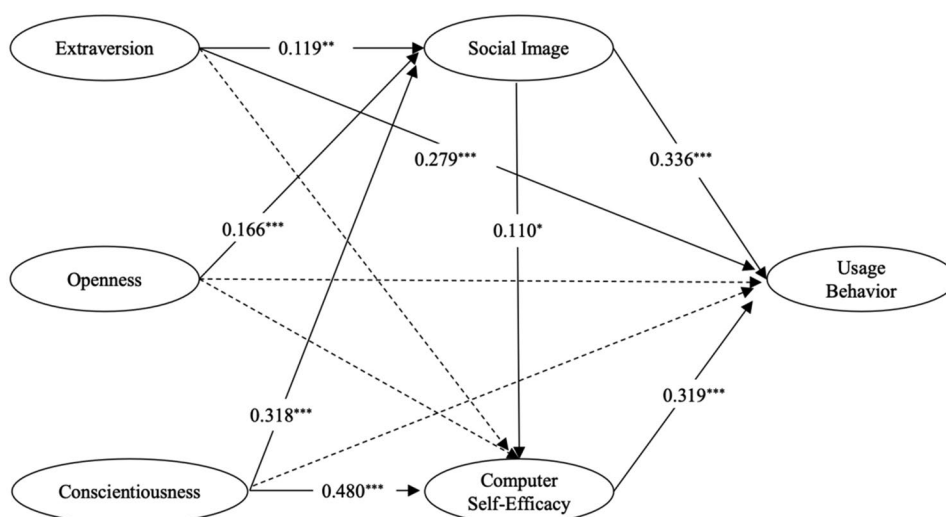
Table 6 presents the results of bootstrapping analyses examining the direct and indirect effects of personality traits on ChatGPT usage through social image and computer self-efficacy. Effects were considered significant when the 95% confidence interval

Table 5. SEM Results.

	S.E.	<i>p</i> value	β
EXV→SIM	0.090	0.005	0.119**
EXV→CSE	0.045	0.390	0.038
EXV→UB	0.072	$p < 0.001$	0.279***
OP→SIM	0.077	$p < 0.001$	0.166***
OP→CSE	0.039	0.118	0.079
OP→UB	0.056	0.155	-0.066
CST→SIM	0.121	$p < 0.001$	0.318***
CST→CSE	0.072	$p < 0.001$	0.480***
CST→UB	0.106	0.218	0.080
SIM→UB	0.033	$p < 0.001$	0.336***
CSE→UB	0.091	$p < 0.001$	0.319***
SIM→CSE	0.022	0.015	0.110*

EXV: Extraversion; OP: Openness; CST: Conscientiousness; SIM: Social image; CSE: Computer self-efficacy; UB: Usage behavior; SEM: Structural equation modeling; S.E.: Standard Error of Mean; β : Standardized regression weights.

* $p < .05$, ** $p < .01$, *** $p < .001$.

**Figure 1.** Hypotheses testing results.

Note. * $p < .05$; ** $p < .01$; *** $p < .001$.

excluded zero at $p < 0.05$ (Preacher & Hayes, 2008). Specifically, the analysis revealed that while extraversion demonstrated a significant positive direct effect (Effect = 0.440, 95% CI = 0.303–0.603), neither openness (Effect = -0.079, 95% CI = -0.183–0.027) nor conscientiousness (Effect = 0.130, 95% CI = -0.102–0.367) showed significant direct relationships with ChatGPT usage behavior. Regarding indirect effects, social image significantly mediated the relationships between ChatGPT usage and three personality traits: extraversion (Effect = 0.063, 95% CI = 0.013–0.114), openness (Effect = 0.067, 95% CI = 0.025–0.116), and conscientiousness (Effect = 0.175, 95% CI = 0.097–0.280). Computer self-efficacy showed significant mediation only for conscientiousness (Effect = 0.250, 95% CI = 0.137–0.408), while its mediating effects were not significant for extraversion (Effect = 0.019, 95% CI = -0.029–0.067) or openness (Effect = 0.030, 95% CI = -0.016–0.082). Furthermore, our results reveal that the three pathways through social image and computer self-efficacy in the serial were statistically significant, and so were the total indirect effects, indicating that social image and computer self-efficacy

Table 6. Bootstrap Analysis of Serial Mediation Effects ($N=784$).

	Effect	BootSE	p value	LLCI	ULCI
Direct effects					
EXV→UB	0.440***	0.077	$p < 0.001$	0.303	0.603
OP→UB	-0.079	0.054	0.142	-0.183	0.027
CST→UB	0.130	0.118	0.258	-0.102	0.367
Indirect effects					
EXV→SIM→UB	0.063*	0.026	0.010	0.013	0.114
OP→SIM→UB	0.067**	0.023	0.001	0.025	0.116
CST→SIM→UB	0.175***	0.047	$p < 0.001$	0.097	0.280
EXV→CSE→UB	0.019	0.024	0.421	-0.029	0.067
OP→CSE→UB	0.030	0.024	0.194	-0.016	0.082
CST→CSE→UB	0.250***	0.071	$p < 0.001$	0.137	0.408
EXV→SIM→CSE→UB	0.007*	0.004	0.032	0	0.017
OP→SIM→CSE→UB	0.007*	0.004	0.024	0.001	0.017
CST→SIM→CSE→UB	0.018*	0.009	0.023	0.003	0.038
Total indirect effects					
EXV→UB	0.089*	0.038	0.028	0.012	0.162
OP→UB	0.104**	0.037	0.004	0.035	0.180
CST→UB	0.443***	0.096	$p < 0.001$	0.284	0.660

EXV: Extraversion; OP: Openness; CST: Conscientiousness; SIM: Social image; CSE: Computer self-efficacy; UB: Usage behavior.
 $*p < .05$, $**p < .01$, $***p < .001$.

sequentially mediated the relationship between the three personality traits and ChatGPT usage behavior.

To sum up, while openness and conscientiousness showed no significant direct effects on ChatGPT usage, they demonstrated significant indirect effects through social image, with conscientiousness also showing effects through computer self-efficacy. Also, although extraversion and openness had no significant indirect effects through computer self-efficacy alone, all three personality traits exhibited significant indirect effects through the sequential mediation of social image and computer self-efficacy. Therefore, our study found hypotheses H3a and H3b were not supported, while supporting H2a, H2b, H2c, H3c, H4b, H4c, and H4d.

Discussion

In our study, empirical evidence supported all proposed hypotheses, except Hypotheses 1b, 1c, 3a and 3b. In terms of relationships between personality traits and ChatGPT usage, the absence of a relationship between openness and ChatGPT usage might be attributed to the standardized responses of ChatGPT, which could potentially fail to meet the preferences of individuals with high openness who tend to value novel and creative discourse. As ChatGPT generates responses based on existing human knowledge, it has inherent limitations in producing original ideas or thinking beyond established frameworks (Barrot, 2024; Biswas, 2023; Peters et al., 2024). Additionally, the lack of relationship between conscientiousness and ChatGPT usage may stem from the characteristic tendency of highly conscientious individuals to deliberate carefully before using new tools. This cautious disposition might generate hesitancy toward ChatGPT usage at its early stages, due to concerns regarding its accuracy and reliability (Kumar et al., 2024). Consequently, individuals with high conscientiousness may gravitate toward established tools that they have already mastered. Further research is needed to elucidate the underlying mechanisms of these relationships. While direct effects from openness to ChatGPT usage and from conscientiousness to ChatGPT usage were

non-significant, both personality traits exhibited significant relationships with ChatGPT usage through sequential mediation *via* social image and computer self-efficacy, consistent with full mediation models (Hayes, 2009; Hayes & Rockwood, 2017). These findings suggest that personality traits alone are not sufficient to drive ChatGPT usage. Rather, social image and computer self-efficacy act as catalysts that bridge personality traits and ChatGPT usage behavior.

Regarding relationships between personality traits and computer self-efficacy, the non-significant relationship between extraversion and computer self-efficacy may be because extraverts typically excel in utilizing visual and auditory cues, such as facial expressions, gestures, tone, and pitch, to communicate effectively in face-to-face interactions. Since ChatGPT interactions are solely text-based, extraverts are unable to leverage their social expressiveness and charisma, resulting in reduced confidence in using ChatGPT effectively (Barrot, 2024; Gurnani & Kaur, 2024; Shahriar & Hayawi, 2023). Furthermore, the non-significant association between openness and computer self-efficacy may be attributed to individuals with high openness typically excelling in creative domains such as design, music, and writing, rather than technical tasks. While these individuals demonstrate proficiency in idea generation and innovative thinking, they might experience lower confidence in utilizing ChatGPT due to its technical nature (Zakiei et al., 2020). In addition, the direct effects of both extraversion and openness on computer self-efficacy, as well as their indirect effects on ChatGPT usage through computer self-efficacy alone, were non-significant. However, when considering the sequential mediating pathway through both social image and computer self-efficacy, the indirect effects of extraversion and openness on ChatGPT usage became significant. This cumulative effect highlights the essential role of social image in mediating the relationship between extraversion and openness on ChatGPT usage.

Among the three personality attributes examined, the path from conscientiousness to ChatGPT usage mediated by social image and computer self-efficacy emerged as the most significant. The finding suggests conscientious individuals more readily perceive ChatGPT as a means to enhance their social status while maintaining high confidence in their abilities to utilize the technology, thus driving their ChatGPT usage behaviors. Conscientious individuals, characterized by their strong sense of obligation toward professional responsibilities, tend to view advanced technologies as valuable instruments for meeting obligations effectively and maintaining their reputation for reliability. In the meantime, their methodical working approach and structured learning style foster greater confidence in technological engagement (Rivers, 2021). Furthermore, the relationship between social image and ChatGPT usage demonstrated greater significance compared to computer self-efficacy. The finding confirms the influential role of social factors, indicating that individuals are motivated to engage with technologies like ChatGPT primarily through their desire to attain and maintain favorable social identities aligned with their technological behaviors (Mehra et al., 2021).

Conclusion

Implications

By integrating TAM3 and the Five Factor Model, this study advances both theories in multiple ways. First, consistent with TAM3, our findings confirm positive relationships

between both social image and computer self-efficacy with AI chat agent usage. Second, we establish a novel association between social image and computer self-efficacy and confirm their positive connections, which was not previously tested in the original TAM3 framework. Third, the study extends TAM3 by incorporating three personality dimensions from the Five Factor Model, creating a more comprehensive model that accounts for both internal psychological factors and external factors that drive AI product usage behaviors. Simultaneously, it extends the Five Factor model by clarifying how distinct personality dimensions relate to social image, computer self-efficacy, and AI chatbot utilization. Fourth, this research broadens TAM3 applicability from organizational technology systems to contemporary AI conversational agents in China, enhancing theoretical generalizability. Last but not least, participants in our study are primarily composed of early users, contributing to strategies that promote AI product usage. According to IDT, early users serve as influential role models who significantly shape the diffusion of innovations throughout society (Qu et al., 2021; Rogers, 2003).

The research offers several practical recommendations for managers seeking to increase AI technology usage. First, AI companies might consider offering users a complimentary trial period, during which they could collect data on the personality traits of users to segment their market more effectively (San-Martin et al., 2020). Recognizing user groups with distinct personality traits can enable AI providers to employ targeted marketing strategies and interventions. Additionally, marketing initiatives should emphasize innovative features of AI products and position them as fashionable lifestyle choices that enhance social identity and self-image of users. Furthermore, AI companies could offer training programs to reduce user anxiety and build confidence.

Limitations and Future Directions

This study has several limitations that need to be addressed in future research. First and foremost, the questionnaire-based approach constrained causal inference, highlighting the importance of experimental research in establishing relationships between variables like social image and computer self-efficacy. Furthermore, self-reported usage data introduced potential subjective bias through overestimation or underestimation of engagement. Future investigations should incorporate objective metrics *via* software tracking to enhance measurement precision and validity (Enrique et al., 2019). In addition, excluding individuals who have not previously used ChatGPT may introduce selection bias and inflate reported efficacy levels. Future research should include user and non-user populations with a comparative analysis to develop a more comprehensive understanding of AI product usage dynamics. Additionally, the predominance of young participants constrained generalizability. Future research should examine diverse age demographics for representative findings. Besides, the pre-survey phase incorporated only eight participants, constituting a limited sample. Subsequent investigations would benefit from expanded pre-survey participation to enhance questionnaire refinement and validation. Moreover, the study overlooked the interaction language of participants with ChatGPT. As generative AI tools like ChatGPT demonstrate varying levels of performance and user experience depending on the language like English versus Mandarin, future research should include this variable in questionnaires. Also, while relationships between studied variables showed statistical significance, their modest effect sizes suggest other important factors remain unexplored. Future research should

examine additional variables, including agreeableness and neuroticism within the Five Factor Model, and perceived usefulness within TAM3. Lastly, the study context within China may limit the generalizability of the result. Replication across diverse cultural contexts would enhance the broader applicability of these findings.

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Data Availability Statement

Data will be made available on request.

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